

MellowYak Phase 16M — Execution and Evidence Report

1. Scope

This is the consolidated English technical report for Phase 16M-C closure on Intel macOS. It records implementation, validation, evidence provenance, limitations, cleanup, and final verdicts. It does not begin Phase 17 and does not convert automated work into human physical evidence.

2. Candidate identity

Field	Value
Version	0.5.0-preview.3
Branch	product/intel-mac-product-lock
Final build commit	71e3955d8a62e546b4a0ac8c34a2fe46f72053b7
Database schema	0011_baseline_lock_and_local_proof
Installed location	/Applications/MellowYak.app
Platform	Intel macOS, x86_64
Default window	1220 × 820

3. Final artifact identity

Artifact	SHA-256
Desktop executable	7d21d73deee3c8606e3cdd0aadbee1512bd07ecf6710e5561c9ccbf476aeb0a5
Embedded engine	ea492adad0b4239b6e4ba5c7ed998d55c4e0dd883d484511202b0b1f49cb8e2e
Browser launcher	97136324f0a487d9fef8eee50341a1b433a05bc4228daae5b1ac19d18ff068fb
Signed DMG	39a4720dcb43ac95eb6ec892f2d4edb169a69c894ff8b516e503ecdd7ae30f73
Deterministic OpenAPI	608ced66dbc65676ab44abae5ed97f070b1ee41af7fc7db127fa35623a66949f

The built and installed desktop/engine hashes matched exactly. The DMG path is apps/desktop/src-tauri/target/release/bundle/dmg/MellowYak_0.5.0-preview.3_x64.dmg .

4. Phase implementation

Notification activation was corrected to revalidate a notification route at activation time. If its project, alert, behavior, change, regression, or gate destination has become stale, the installed UI opens the safe Alerts route. A React regression test proves the fallback and prevents the former stale-context navigation behavior.

Version metadata was synchronized across the desktop package, Tauri configuration, Rust lockfile, Python engine metadata, OpenAPI metadata, and packaged validators. The final candidate was rebuilt from the exact recorded commit.

5. Finder Alias relocation and reconnect

A disposable project was relocated outside the repository. Through the installed application and native folder chooser:

- The project was disconnected.
- Reconnect selected a genuine Finder Alias.
- Canonical resolution found the relocated directory.
- The original project ID 082d9fa2-95aa-43c3-9178-21ef9a71ba37 and its history were preserved.
- The database contained exactly one project record.
- A foreign fixture was then selected and was rejected with: "Project identity does not match. Existing history was not attached and no source was changed."
- The valid Alias was selected again and the project returned to Connected.

This closes the automatable and functional axes for P15-PHYS-120 without claiming a human physical click.

6. Quiet Mode

Quiet Mode was enabled for one hour in the installed application. A real, eligible warning alert was inserted into the isolated acceptance database. The alert remained visible and actionable inside Alerts while native delivery was suppressed. Quiet Mode was ended, its persisted state returned to inactive, and a second fresh eligible alert exercised the post-quiet path.

macOS Screen Recording permission was unavailable, so no new full-screen capture is presented as physical evidence. The functional axis passes; the human-observed native before/after banner sequence remains pending. Quiet Mode was restored to off and the synthetic alerts were removed during cleanup.

7. Source verification matrix

Check	Result
Python tests	PASS — 210 passed
React/Vitest	PASS — 30 passed
TypeScript	PASS
Vite production build	PASS
GUI translation-key-only check	PASS
English/Hebrew catalog parity	PASS
Hebrew RTL behavior	PASS
Ruff check and format	PASS
Cargo format and check	PASS
Migration matrix, empty and 0001–0010 inputs to 0011	PASS
OpenAPI export repeated byte-identically	PASS

The single Python warning was a third-party Starlette deprecation warning and did not change the result.

8. Packaged verification matrix

Validator	Result
Phase 8 packaged validation	Completed; evidence output retained locally
Phase 9 packaged validation	Completed; evidence output retained locally
Phase 10 product-truth/self-test	VERIFIED_WORKING
Phase 11M macOS package	VERIFIED_WORKING after outer DMG signing correction
Phase 12 safe apply	VERIFIED_WORKING
Phase 13 orchestration	VERIFIED_WORKING
Phase 15 product lock	VERIFIED_WORKING
Native lifecycle	VERIFIED_WORKING
Updater E2E	VERIFIED_WORKING with ephemeral acceptance keys
External product network	None observed

Phase 12 included a successful apply, a forced post-check failure, byte-identical rollback, and stale-source rejection. Native lifecycle included clean launch, single-instance focus, one engine, supervised engine restart, explicit exit, and owned-child cleanup. Updater E2E rejected tampered, wrong-key, and incomplete packages while preserving data.

9. DMG correction

The first 0.5.0-preview.3 DMG run exposed `dmg_codesign_structure_valid=false`. The outer DMG was then ad-hoc signed and the Phase 11M validator was rerun. The corrected final DMG passed signature structure, read-only mount, exact bundle contents, and clean detach. Only the corrected final hash is listed as the candidate artifact.

10. Evidence model

All reports use schema `mellowyak.phase16m.acceptance-evidence.v2` with five independent axes:

- `native_automation_status`
- `human_physical_status`
- `visual_status`
- `functional_status`
- `cleanup_status`

The authoritative per-test matrix is in `PHASE_16M_PHYSICAL_ACCEPTANCE.md` and the machine-readable equivalent is in `PHASE_16M_EVIDENCE_MANIFEST.json`.

11. Visual evidence rules

- Files under `images/native` and `images/product` are canonical captures from the installed acceptance work.
- Files under `images/native/notifications` are genuine Notification Center banner captures.
- Files under `images/deterministic` are labeled `DETERMINISTIC_PRODUCT_VIEW — NOT HUMAN PHYSICAL EVIDENCE`.
- The same canonical image may appear in both PDFs.
- Accessibility automation is evidence for native automation only.
- No screenshot is promoted to a stronger evidence class than its origin supports.

12. Privacy remediation

Two original System Settings screenshots exposed unrelated account/sidebar context. They were moved outside the repository and replaced with canonical crops containing only the MellowYak notification controls. The crop operation used local macOS image tooling and did not synthesize UI. The delivery directory was checked to exclude customer source, credentials, databases, browser profiles, support bundles, private full-desktop captures, and normal product data.

13. Localization

All GUI copy remains translation-key driven. English is the base catalog; Hebrew contains matching keys and uses RTL. Source checks and React tests passed. The documentation screenshots are English-only by design; localization completeness is verified in code so future catalogs such as Russian can be added without hardcoded component text.

14. Human-only boundary

The following remain `HUMAN_PHYSICAL_NOT_RUN` where a real physical boundary is required:

- logout/login with startup enabled and disabled;
- lock/unlock;
- idle and queued sleep/wake;
- complete macOS restart;
- real battery-power transition;
- physical delivered-notification activation for the stale-destination scenario;
- human-observed native banner comparison across Quiet Mode.

These are intentionally not replaced by Accessibility automation or deterministic images.

15. Distribution boundary

The app and DMG are ad-hoc signed local technical-preview artifacts. Developer ID signing, notarization, stapling, trusted public distribution, and production updater trust are not configured. Gatekeeper correctly rejected a quarantined ad-hoc build; no bypass is documented.

16. Cleanup and restoration

The acceptance configuration is restored to Start at Login off, close-to-tray on, Quiet Mode off, and Activity Mode normal. Acceptance-lab environment variables are unset. Disposable alerts, aliases, test roots, previous installer backup, and isolated acceptance data are removed only after evidence generation. Normal user data is untouched. No app, engine, helper, or mounted DMG is intentionally left after final closure cleanup.

17. Git boundary

The implementation commit is local. Documentation and canonical evidence are committed separately after PDF validation. Generated app bundles, DMGs, runtime data, build manifests, Finder Alias artifacts, databases, and private acceptance state are excluded. Nothing is pushed. No `physical-accepted` tag is created.

18. Final verdicts

- Product: `INTEL_MAC_TECHNICAL_PREVIEW_ACCEPTED_WITH_LIMITS`.
- Human physical: `HUMAN_PHYSICAL_ACCEPTANCE_PENDING`.
- Distribution: `PUBLIC_MAC_DISTRIBUTION_BLOCKED`.
- Local tag: not created at this closure because the mandatory physical boundary is incomplete.
- Exact next authorized phase: finish the remaining Phase 16M human physical checks against this exact candidate; Phase 17 is not authorized.

19. Complete test-axis matrix

Expected and actual details for every row are maintained in `PHASE_16M_PHYSICAL_ACCEPTANCE.md`; the same machine-readable axes and evidence paths are in the v2 manifest.

Test ID	Native automation	Human physical	Visual	Functional	Cleanup
P15-PHYS-101	NATIVE_AUTOMATION_PASS	HUMAN_PHYSICAL_NOT_RUN	VISUAL_PASS	PASS	PASS
P15-PHYS-102	NATIVE_AUTOMATION_PASS	HUMAN_PHYSICAL_NOT_RUN	NOT_RUN	PASS	PASS
P15-PHYS-103	NATIVE_AUTOMATION_PASS	HUMAN_PHYSICAL_NOT_RUN	NO_VISUAL_EVIDENCE_REQUIRED	PASS	PASS
P15-PHYS-104	NATIVE_AUTOMATION_PASS	HUMAN_PHYSICAL_NOT_RUN	NOT_RUN	PASS	PASS
P15-PHYS-105	NATIVE_AUTOMATION_PASS	HUMAN_PHYSICAL_NOT_RUN	NOT_RUN	PASS	PASS
P15-PHYS-106	NATIVE_AUTOMATION_PASS	HUMAN_PHYSICAL_NOT_RUN	NOT_RUN	PASS	PASS
P15-PHYS-107	NATIVE_AUTOMATION_PASS	HUMAN_PHYSICAL_NOT_RUN	NOT_RUN	PASS	PASS
P15-PHYS-108	NATIVE_AUTOMATION_PASS	HUMAN_PHYSICAL_NOT_RUN	VISUAL_PASS	PASS	PASS

P15-PHYS-109	NATIVE_AUTOMATION_PASS	HUMAN_PHYSICAL_NOT_RUN	VISUAL_PASS	PASS	PASS
P15-PHYS-110	NATIVE_AUTOMATION_PASS	HUMAN_PHYSICAL_NOT_RUN	VISUAL_PASS	PASS	PASS
P15-PHYS-111	NATIVE_AUTOMATION_PASS	HUMAN_PHYSICAL_NOT_RUN	VISUAL_PASS	PASS	PASS
P15-PHYS-112	BLOCKED	HUMAN_PHYSICAL_NOT_RUN	NOT_RUN	PASS	PASS
P15-PHYS-113	NATIVE_AUTOMATION_PASS	HUMAN_PHYSICAL_NOT_RUN	NOT_RUN	PASS	PASS
P15-PHYS-114	NOT_RUN	HUMAN_PHYSICAL_NOT_RUN	NOT_RUN	NOT_RUN	PASS
P15-PHYS-115	NATIVE_AUTOMATION_PASS	HUMAN_PHYSICAL_NOT_RUN	NOT_RUN	PASS	PASS
P15-PHYS-115B	NOT_RUN	HUMAN_PHYSICAL_NOT_RUN	NOT_RUN	NOT_RUN	PASS
P15-PHYS-116	NOT_RUN	HUMAN_PHYSICAL_NOT_RUN	NOT_RUN	NOT_RUN	PASS
P15-PHYS-117	NOT_RUN	HUMAN_PHYSICAL_NOT_RUN	NOT_RUN	NOT_RUN	PASS
P15-PHYS-118	NOT_RUN	HUMAN_PHYSICAL_NOT_RUN	NOT_RUN	NOT_RUN	PASS
P15-PHYS-119	NATIVE_AUTOMATION_PASS	HUMAN_PHYSICAL_NOT_RUN	VISUAL_PASS	PASS	PASS
P15-PHYS-120	NATIVE_AUTOMATION_PASS	HUMAN_PHYSICAL_NOT_RUN	VISUAL_PASS	PASS	PASS
P15-PHYS-121	NATIVE_AUTOMATION_PASS	HUMAN_PHYSICAL_NOT_RUN	NOT_RUN	PASS	PASS
P15-PHYS-122	NATIVE_AUTOMATION_PASS	HUMAN_PHYSICAL_NOT_RUN	NOT_RUN	PASS	PASS
P15-PHYS-123	NATIVE_AUTOMATION_PASS	HUMAN_PHYSICAL_NOT_RUN	NOT_RUN	PASS	PASS
P15-PHYS-124	NATIVE_AUTOMATION_PASS	HUMAN_PHYSICAL_NOT_RUN	NOT_RUN	PASS	PASS
P15-PHYS-125	NATIVE_AUTOMATION_PASS	HUMAN_PHYSICAL_NOT_RUN	NOT_RUN	PASS	PASS
P15-PHYS-126	NOT_RUN	HUMAN_PHYSICAL_NOT_RUN	NOT_RUN	NOT_RUN	PASS
P15-PHYS-127	BLOCKED	HUMAN_PHYSICAL_NOT_RUN	VISUAL_PASS	PASS	PASS
P15-PHYS-128	NOT_RUN	HUMAN_PHYSICAL_NOT_RUN	NOT_RUN	NOT_RUN	PASS

Expected, actual, evidence, cleanup, and remaining action

+| Test ID | Native automation | Human physical | Visual | Functional | Expected | Actual | Evidence | Cleanup | Remaining human action |

P15-PHYS-101	NATIVE_AUTOMATION_PASS	HUMAN_PHYSICAL_NOT_RUN	VISUAL_PASS	PASS	One translated menu and truthful counters	Native menu inspected; one app/engine	Native tray capture and lifecycle log	P,
P15-PHYS-102	NATIVE_AUTOMATION_PASS	HUMAN_PHYSICAL_NOT_RUN	NOT_RUN	PASS	Existing window focuses safely	Same window/process restored	Installed lifecycle log	P,
P15-PHYS-103	NATIVE_AUTOMATION_PASS	HUMAN_PHYSICAL_NOT_RUN	NO_VISUAL_EVIDENCE_REQUIRED	PASS	No duplicate app or engine	Second instance exited; one app/engine	Lifecycle validator	P,
P15-PHYS-104	NATIVE_AUTOMATION_PASS	HUMAN_PHYSICAL_NOT_RUN	NOT_RUN	PASS	Window hides; monitoring continues	Existing identity remained active	Lifecycle validator	P,
P15-PHYS-105	NATIVE_AUTOMATION_PASS	HUMAN_PHYSICAL_NOT_RUN	NOT_RUN	PASS	Close follows explicit exit policy	Policy matched and setting restored	Lifecycle validator	P,
P15-PHYS-106	NATIVE_AUTOMATION_PASS	HUMAN_PHYSICAL_NOT_RUN	NOT_RUN	PASS	All owned processes exit	Desktop, engine and children exited	Process evidence	P,
P15-PHYS-107	NATIVE_AUTOMATION_PASS	HUMAN_PHYSICAL_NOT_RUN	NOT_RUN	PASS	Cmd+Q/menu Quit cleans all children	Shutdown defect fixed; cleanup passed	Process evidence	P,
P15-PHYS-108	NATIVE_AUTOMATION_PASS	HUMAN_PHYSICAL_NOT_RUN	VISUAL_PASS	PASS	OS and product permission agree	MellowYak permission enabled	Privacy-safe Settings crop	P,
P15-PHYS-109	NATIVE_AUTOMATION_PASS	HUMAN_PHYSICAL_NOT_RUN	VISUAL_PASS	PASS	In-app evidence persists; no false claim	Denied state observed and restored	Privacy-safe Settings crop	P,
P15-PHYS-110	NATIVE_AUTOMATION_PASS	HUMAN_PHYSICAL_NOT_RUN	VISUAL_PASS	PASS	Exactly one redacted native banner	Seven genuine banner scenarios captured	Native notification images	P,
P15-PHYS-111	NATIVE_AUTOMATION_PASS	HUMAN_PHYSICAL_NOT_RUN	VISUAL_PASS	PASS	Existing app opens safe destination	Alerts destination opened without duplicate	Installed product capture	P,
P15-PHYS-112	BLOCKED	HUMAN_PHYSICAL_NOT_RUN	NOT_RUN	PASS	Revalidate and fall back to Alerts	Click-time code/test passes; real stale delivered-banner activation not completed	Source test and build commit	P,

P15-PHYS-113	NATIVE_AUTOMATION_PASS	HUMAN_PHYSICAL_NOT_RUN	NOT_RUN	PASS	Exactly one valid registration	One LaunchAgent registered	LaunchAgent inspection	P,
P15-PHYS-114	NOT_RUN	HUMAN_PHYSICAL_NOT_RUN	NOT_RUN	NOT_RUN	One app/tray/engine after login	Real login boundary not performed	None	P,
P15-PHYS-115	NATIVE_AUTOMATION_PASS	HUMAN_PHYSICAL_NOT_RUN	NOT_RUN	PASS	Registration removed	LaunchAgent removed and setting restored	LaunchAgent inspection	P,
P15-PHYS-115B	NOT_RUN	HUMAN_PHYSICAL_NOT_RUN	NOT_RUN	NOT_RUN	No automatic process after login	Real login boundary not performed	None	P,
P15-PHYS-116	NOT_RUN	HUMAN_PHYSICAL_NOT_RUN	NOT_RUN	NOT_RUN	One coherent process set after unlock	Authentication boundary not performed	None	P,
P15-PHYS-117	NOT_RUN	HUMAN_PHYSICAL_NOT_RUN	NOT_RUN	NOT_RUN	Watcher resumes without duplicate	Physical sleep cycle not performed	None	P,
P15-PHYS-118	NOT_RUN	HUMAN_PHYSICAL_NOT_RUN	NOT_RUN	NOT_RUN	Queued work resumes or invalidates safely	Controlled sleep cycle not performed	None	P,
P15-PHYS-119	NATIVE_AUTOMATION_PASS	HUMAN_PHYSICAL_NOT_RUN	VISUAL_PASS	PASS	Alias resolves to canonical project root	Native chooser resolved disposable Alias	Canonical-resolution capture	P,
P15-PHYS-120	NATIVE_AUTOMATION_PASS	HUMAN_PHYSICAL_NOT_RUN	VISUAL_PASS	PASS	Same identity/history; mismatch rejected	Same ID retained; one record; foreign fixture rejected; valid source restored	Installed AX, database evidence, canonical/deterministic images	P,
P15-PHYS-121	NATIVE_AUTOMATION_PASS	HUMAN_PHYSICAL_NOT_RUN	NOT_RUN	PASS	Exact app installs from read-only DMG	Mounted, verified, installed and detached	Phase 11M validator	P,
P15-PHYS-122	NATIVE_AUTOMATION_PASS	HUMAN_PHYSICAL_NOT_RUN	NOT_RUN	PASS	Cold launch produces one app/engine	Installed hashes matched; lifecycle passed	Hash and process evidence	P,
P15-PHYS-123	NATIVE_AUTOMATION_PASS	HUMAN_PHYSICAL_NOT_RUN	NOT_RUN	PASS	No duplicate identity or process	Reinstall passed with one app/engine	Package/lifecycle evidence	P,
P15-PHYS-124	NATIVE_AUTOMATION_PASS	HUMAN_PHYSICAL_NOT_RUN	NOT_RUN	PASS	No processes/startup item/source mutation	App removal preserved isolated data	Package evidence	P,
P15-PHYS-125	NATIVE_AUTOMATION_PASS	HUMAN_PHYSICAL_NOT_RUN	NOT_RUN	PASS	Retained data opens without corruption	Reinstall passed; one process set	Package evidence	P,
P15-PHYS-126	NOT_RUN	HUMAN_PHYSICAL_NOT_RUN	NOT_RUN	NOT_RUN	Documented startup; no duplicate recovery	Real restart not performed	None	P,
P15-PHYS-127	BLOCKED	HUMAN_PHYSICAL_NOT_RUN	VISUAL_PASS	PASS	In-app alert persists; native delivery suppressed, then resumes	Real alert persisted while quiet and state restored; post-quiet native banner was not physically observed	Installed DB/UI evidence plus deterministic Settings view	P,
P15-PHYS-128	NOT_RUN	HUMAN_PHYSICAL_NOT_RUN	NOT_RUN	NOT_RUN	Optional work defers on real battery state	Host remained on AC; no simulation claimed	Power-state observation	P,